SEAT NO: [redacted]

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(Data Science)

SPRING SEMESTER EXAMINATIONS 2026

Time : 3 Hours | **Batch 2023** | **Dated : 08-JUN-26**

Max Marks : 60

Data Mining - CT-356

Note: Attempt all questions. Answer all parts of a question in a row. State your assumptions clearly. Draw diagram where necessary.

Q1. **[CLO 1: 12 Marks]**

a) ***Explain*** the following with proper reasoning and examples. **[8]**

1) Differentiate between Centroid and Medoid
2) How do rare labels affect the performance of Data Mining Models?
3) What is the impact of low value of minimum support count?
4) Why are foundational clustering algorithms considered to be semi-supervised learning?

b) A fraud detection system is developed using a Support Vector Classifier to distinguish between fraudulent (0) and legitimate (1) transactions. The classifier correctly predicts 95 out of 100 transactions, yielding an accuracy of 95%. ***Explain*** whether this model is truly performing well with suitable reasoning, considering the nature of the dataset and appropriate evaluation metrics. **[4]**

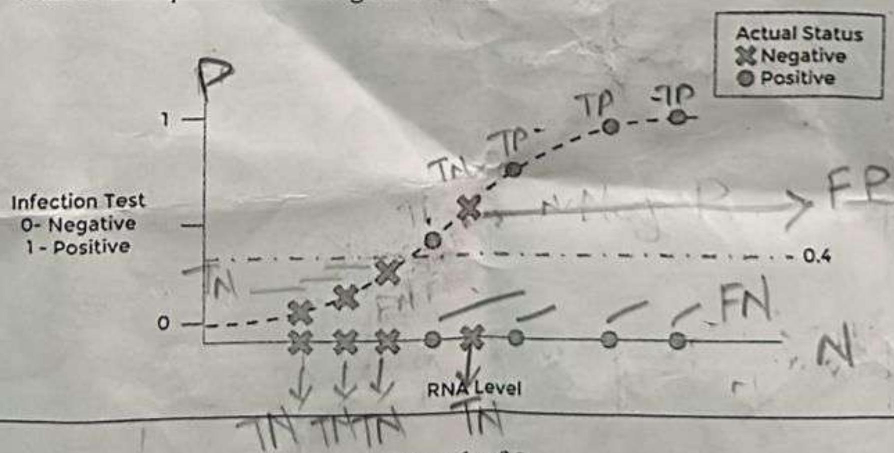
Q2. **[CLO 1: 12 Marks]**

a) A data mining model fitted on a sentiment analysis dataset. ***Summarize*** the model's performance by applying the following evaluation metrics: Confusion Matrix, Accuracy, Support, Weighted Precision, Weighted Recall, and F-Measure. Consider the following scenario; **[8]**

- **Negative:** 93 were correctly identified, 5 were misidentified as neutral, 12 were misclassified as positive
- **Neutral:** 35 were correctly identified, 6 were misidentified as negative, 9 were misclassified as positive
- **Positive:** 85 were correctly identified, 11 were misidentified as negative, 9 were misclassified as neither positive nor negative

b) Given a logistic regression graph used to predict whether a patient is infected (Positive) or not infected (Negative) based on RNA level. ***Interpret*** the results and answer the following questions. **[4]**

1) Find out the TP, TN, FP, FN
2) What is the impact of decreasing cut-off value



Q3. Using Naïve Bayes algorithm to the following table and perform ***inference*** on the class outcome of play for Outlook=Overcast, Temp=Cool, Humidity=High, and Windy=Strong. **[CLO 2: 12 Marks]**

Sr. #	Outlook	Temp	Humidity	Windy	Play
1	Sunny	Hot	High	Normal	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Normal	Yes
4	Rainy	Mild	High	Normal	Yes
5	Rainy	Cool	Normal	Normal	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Normal	No
9	Sunny	Cool	Normal	Normal	Yes
10	Rainy	Mild	Normal	Normal	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Normal	Yes
14	Rainy	Mild	High	Strong	No
15	**Overcast**	**Cool**	**High**	**Strong**	**?**

Q4. Data have been downloaded from the Daraz 11-11 sale packages catalog by some business experts to learn the pattern of frequency of items. There are 9 different transactions (order 1 to 9) and each transactions have minimum 2 and maximum four items. Here 5 different kind of items are available. For simplicity Our Team lead used short names of items (I1- I 5) rather than full descriptive name (example I1= Hair dryer). Consider Minimum threshold support as 22.22% and confidence as 77.77%. ***Discover*** frequent k items by applying the association rule-based mining technique that is FP-Growth algorithm. Show all your *steps* with detail graphical and tabular format. **[CLO 2: 12 Marks]**

Catalogue Item	List of Item in Package
Package 01	I1, I2, I5
Package 02	I2, I4
Package 03	I2, I3
Package 04	I1, I2, I4
Package 05	I1, I3
Package 06	I2, I3
Package 07	I1, I3
Package 08	I1, I2, I3, I5
Package 09	I1, I2, I3

Q5. Apply the DBSCAN algorithm to the given data points to form clusters using **minPts = 4** and $\boldsymbol{\varepsilon = 1.9}$. Use the Minkowski distance with $r = 2$ (l_2 *norm*) to calculate the distance between each points. ***Categorize*** the core, border and noise points, and illustrate the resulting clusters with a graph. **[CLO 2: 12 Marks]**

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
3,7	4,6	5,5	6,4	7,3	6,2	7,2	8,4	3,3	2,6	3,5	2,4